Sensitivity, Informativeness, and Misspecification in GMM Estimation

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Motivation

- Sensitivity to model specification is central to assessing the **robustness** of empirical conclusions.
- Reduced form: a rich, standardized toolkit to check omitted variable bias formula. (Rosenbaum and Rubin 1983; Imbens 2003; Oster 2019; Cinelli and Hazlett 2020)
- Structural estimators: far less standardized.
    - → the map from moment conditions to parameters is implicit and nonlinear;
    - → misspecification is prevalent in applied work.

Sensitivity is a local robustness diagnostic

- **Sensitivity** (Andrews, Gentzkow, and Shapiro 2017, AGS): how $\hat{\theta}$ responds to each estimation moment.
- A local deviation of the moments, evaluated at correct specification.
- But in their analysis of Berry, Levinsohn, and Pakes (1995) (BLP):
    - → the J-test rejects strongly
    - → the moments are globally (fixed) misspecified.

Sensitivity diagnostics meet misspecification

- The J-test rejects in practice:
  - → Berry, Levinsohn, and Pakes (1995) (BLP): $J = 345$ (df 25).
  - → Blundell, Pistaferri, and Preston (2008) (BPP): $J = 539$ (df 290).
- Misspecification $\Rightarrow$ $\hat{\theta}$ converges to the pseudo-true value θ_0, the unique minimizer of the GMM criterion, with

$$g = \mathbb{E}[g(X_i, \theta_0)] \neq 0.$$

- **Extend AGS**: sensitivity as a local deviation around the globally misspecified θ_0

Setup

- Sample moments $\hat{g}(\theta)$.
- Joint asymptotic distribution of the estimator and the moments:

$$\sqrt{n}\begin{pmatrix}\hat{\theta}-\theta_0\\ \hat{g}(\theta_0)-g\end{pmatrix} \rightsquigarrow \begin{pmatrix}\tilde{\theta}\\ \tilde{g}\end{pmatrix} \sim \mathcal{N}\left(0,\ \begin{pmatrix}\sigma_{\theta\theta} & \sigma_{\theta g}\\ \sigma_{g\theta} & \sigma_{gg}\end{pmatrix}\right).$$

DEFINITION 1 *(misspecification-robust sensitivity, MRS)*

$$\Lambda = \sigma_{\theta g}\,\sigma_{gg}^{-1} \qquad (p \times q).$$

In the limit, $\mathbb{E}[\tilde{\theta} \mid \tilde{g}] = \Lambda\,\tilde{g}$: the regression coefficient of the estimator on the moments.

Correct specification: $\quad \Lambda_{AGS} = -(G'WG)^{-1}G'W$.

Sensitivity requires the influence function

- Asymptotically linear representation:

$$\sqrt{n}(\hat{\theta} - \theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi(X_i) + o_p(1), \qquad \nu_i = g(X_i, \theta_0) - g.$$

- Then $\Lambda = \mathbb{E}[\psi\nu'] \, \mathbb{E}[\nu\nu']^{-1}$: we need $\psi(X_i)$ at the pseudo-true value.
- Correct specification: $\psi_i = \Lambda_{AGS} \, \nu_i$.
- Misspecification ($g \neq 0$): sampling variation in the Jacobian $\widehat{G}$ and the estimated weight $\widehat{W}(\widehat{\phi})$, with $\widehat{\phi}$ a first-step estimator, enters ψ at first order, through influence functions γ_i, ω_i, ψ^{ϕ}.

Proposition 1 derives ψ and these channels.

Channel decomposition

PROPOSITION 1 *(decomposition of GMM influence functions)*

One-step GMM with estimated weight $\widehat{W}$:

$$\psi^{1s,W}(X_i) = -A^{-1}\Big[\underbrace{G'W(g_i - g)}_{\text{moment}} + \underbrace{(g'W \otimes I_p)\,\gamma_i}_{\text{Jacobian}} + \underbrace{(g' \otimes G')\,\omega_i}_{\text{weight matrix}}\Big].$$

Two-step adds $B\psi^{\phi}(X_i)$ inside the bracket; iterated replaces A^{-1} by $(A+B)^{-1}$.

DEFINITION 2 *(informativeness)*

$$\Delta_k = \frac{\sigma_{\theta_k g}\,\sigma_{gg}^{-1}\,\sigma_{g\theta_k}}{\sigma_{\theta_k\theta_k}} = R^2 \text{ of } \hat{\theta}_k \text{ on the moments } \in [0,1].$$

Curvature A and weight-Jacobian B

The bracket in Proposition 1 is scaled by the curvature A; the first step and iteration enter through B.

$$A = \underbrace{G'WG}_{\text{standard curvature}} + \underbrace{H}_{\text{misspecification}}, \qquad H = (g'W \otimes I_p)\,R, \quad R = \mathbb{E}\big[\partial_{\theta'} \operatorname{vec}(G(X_i, \theta_0)')\big].$$

$$B = (g' \otimes G')\,S, \qquad S = \begin{cases} \partial_{\phi'} \operatorname{vec} W(\phi_0) & \text{(two-step)} \\ \partial_{\theta'} \operatorname{vec} W(\theta_0) & \text{(iterated).} \end{cases}$$

- Under correct specification ($g = 0$): $H = 0$ and $B = 0$, so $A = G'WG$ – the standard GMM curvature, and the first-step/iteration channel vanishes.

Informativeness, and when $\Delta_k < 1$

- We interpret informativeness as **structural efficiency**: the proportion of the asymptotic variance explained by the moments.
- Extends Andrews, Gentzkow, and Shapiro (2020)
  - → they derive the informativeness of descriptive statistics for correctly specified GMM estimator
  - → replacing descriptive statistics by g, $\Delta = 1$ under correct specification
- $\Delta_k < 1$ when the Jacobian channel, the weight-matrix channel, or the preliminary-estimator channel matter.
- Δ_k vs. the J-test.

Optimal minimum distance

(Proposition 2 in the paper)

- Minimum distance, $g(X_i, \theta) = m_i - f(\theta)$, with the optimal weight $\widehat{W} = \widehat{V}^{-1}$.
- Jacobian is nonrandom and there is no first step.
- Only the weight-matrix channel survives: we show

$$\psi_i = M_\nu \nu_i \, (1 - \nu_i' W g).$$

- Under Gaussian moments, we show:

$$\Delta_k = \frac{1}{1 + g' W g}.$$

- The optimal weight buys efficiency at the price of informativeness.

The weighting ladder

$V = I_q$, Gaussian, common misspecification $g_k = \delta$. Same population weight; differ in how much is estimated:

$$\underbrace{\Delta_k^{OMD} = \frac{1}{1 + q\delta^2}}_{\text{optimal weight}} \qquad \underbrace{\Delta_k^{DWMD} = \frac{1}{1 + 2\delta^2}}_{\text{diagonal}} \qquad \underbrace{\Delta_k^{EWMD} = 1}_{\text{identity}}.$$

- OMD channel grows with q; in BPP, $q = 325$.
- Asymptotic counterpart of Altonji and Segal (1996).

Sensitivity to the first step, and iteration

PROPOSITION 3 *(sensitivity to the first-step estimator)*

$$\Lambda_\phi = \operatorname{plim} \frac{\partial \hat{\theta}(\widehat{\phi})}{\partial \widehat{\phi}'} = -A^{-1}B.$$

Iterated GMM as a fixed point

PROPOSITION 4 *(geometric convergence of iterated GMM)*

For every $\tilde{\rho} \in (\rho(-A^{-1}B), 1)$, there is $C_{\tilde{\rho}}$ with

$$\left\| \psi^{ss}(X_i) - \psi^{it}(X_i) \right\|_{L^2(P)} \leq C_{\tilde{\rho}}\, \tilde{\rho}^{\,s-1}.$$

Hence $\mathbf{Var}(\psi^{ss}) \to \mathbf{Var}(\psi^{it})$ and $\Delta_k^{ss} \to \Delta_k^{it}$ at the same rate.

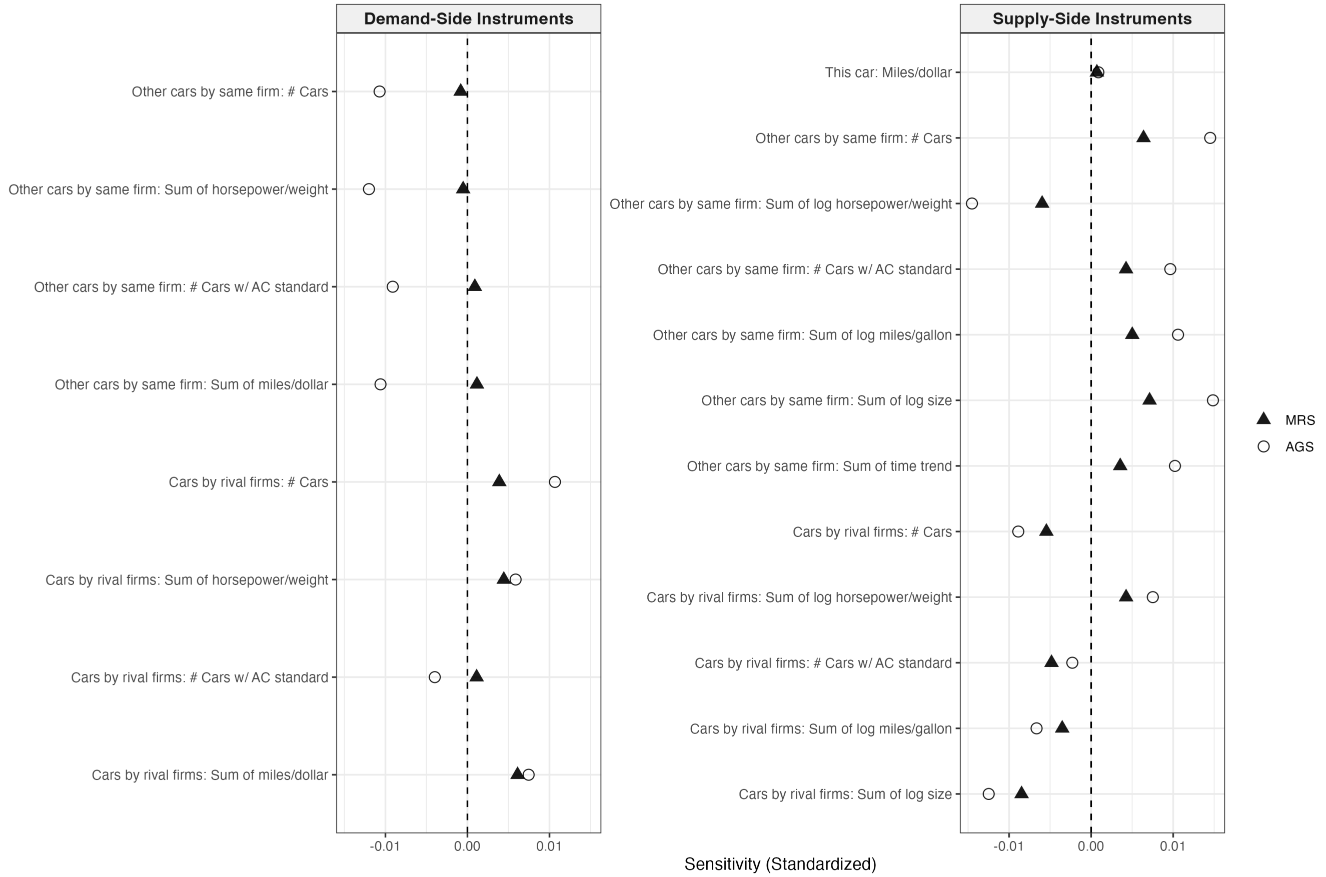
- Hansen and Lee (2021) show convergence of the iterated estimand/estimator.
- This proposition shows the convergence holds for the influence function and the variance as well.

Three applications

	Jacobian γ_i	weight channel	iteration
BLP	large	active	yes
BPP	$= 0$	active	none
AJRY	modest	active	yes

- BLP: misspecification reorders *sensitivity*.
- BPP: the *choice of weight* as an efficiency–informativeness trade.
- AJRY: J and Δ *disagree*.

BLP: sensitivity reorders



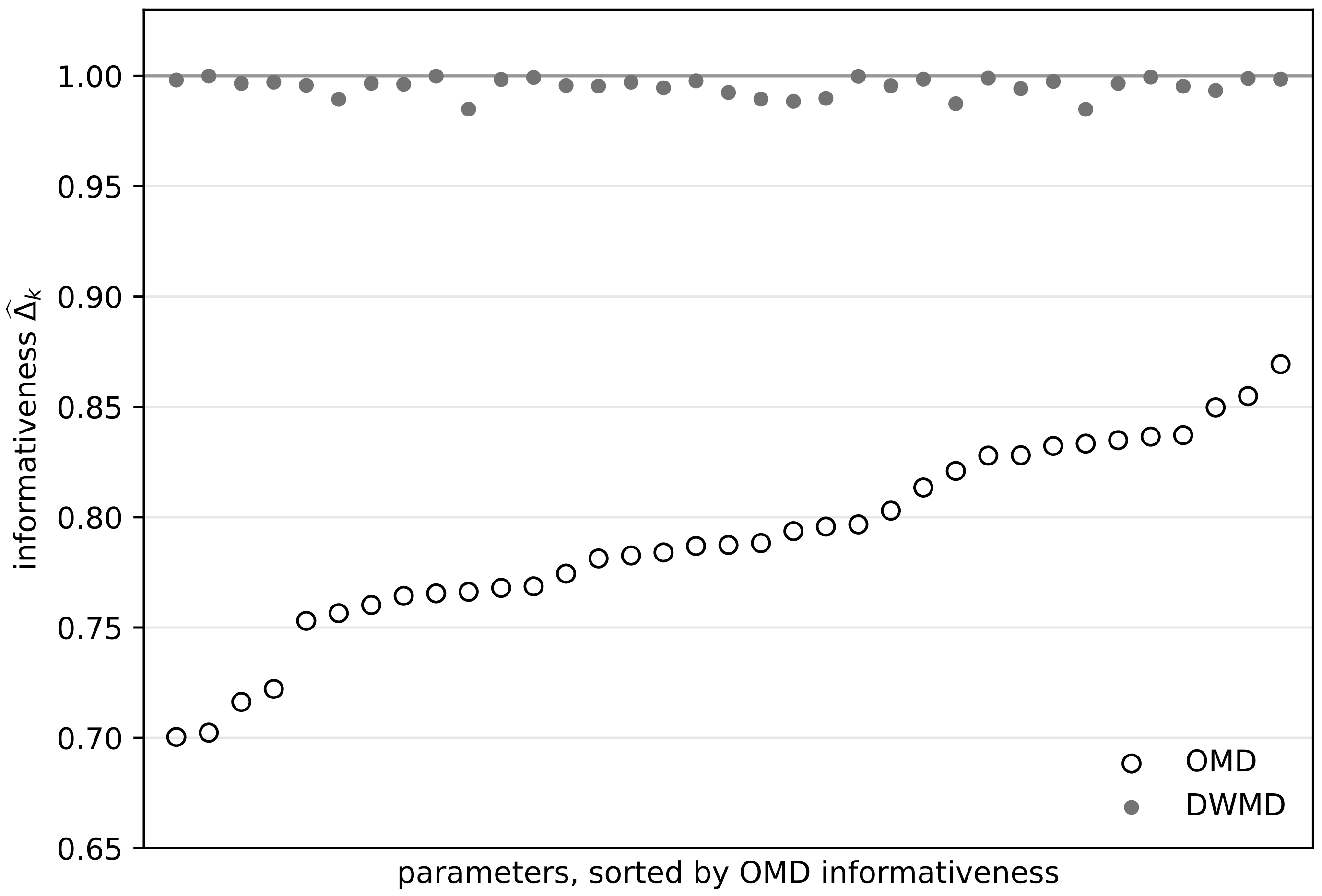
- AGS (hollow) vs. MRS (filled) for the average markup.
- $\widehat{\Delta}_{markup} = 0.56, \; J = 345$ (p-value < 0.001).

BPP: three weights, same moments

35 parameters, $q = 325$, $n = 1{,}765$, $J = 538.7$ (df 290).

	$\widehat{\phi}$ (permanent)	SE	$\widehat{\Delta}$	$\widehat{\psi}$ (transitory)	SE	$\widehat{\Delta}$
OMD	0.33	0.043	0.79	0.07	0.039	0.79
DWMD	0.68	0.113	0.99	0.03	0.043	0.99
EWMD	0.61	0.109	1.00	0.00	0.049	1.00

BPP: efficiency vs. informativeness



OMD 0.70–0.87; DWMD ≈ 0.99; EWMD $= 1$. SE(ϕ): $0.043 \to 0.113$.

AJRY: income and democracy

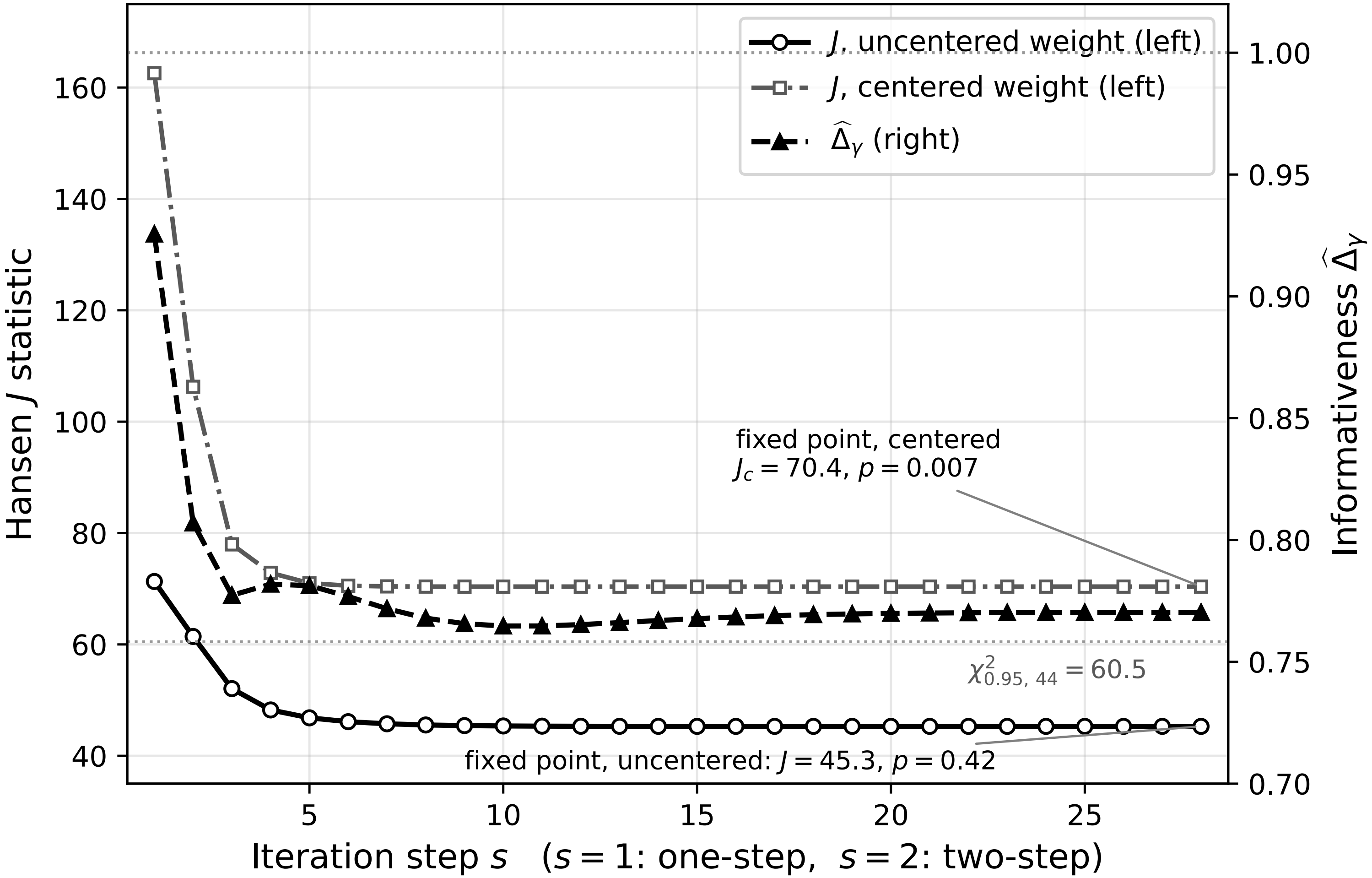
Difference GMM, $n = 127$ countries, df 44; coefficient γ.

estimator	$\widehat{\gamma}$	J	p	$\widehat{\Delta}_\gamma$
one-step (AB weight)	-0.129	71.3	0.006	0.93
two-step efficient	-0.012	61.4	0.04	0.81
iterated fixed point	-0.009	45.3	0.42	0.77

- Efficient weight lowers $\widehat{\Delta}_\gamma$: $0.93 \to 0.81 \to 0.77$.
- J stops rejecting (uncentered efficient weight).
- Sensitive to the choice of uncentered/centered efficient weight:

$$J = 45.3\,(p{=}0.42) \;\; \text{vs.} \;\; J_c = 70.4\,(p{=}0.007), \qquad J = J_c/(1 + J_c/n).$$

AJRY: J turns on a centering convention; Δ does not



$$\widehat{\Delta}_\gamma = 0.77,\ J = 45.3,\ J_c = 70.4.$$

J and Δ are complements

	J under iteration	$\widehat{\Delta}$ under iteration	
BLP	flat at 345	$0.56 \rightarrow 0.24$	J silent, Δ falls
BPP	no iteration	0.79 / 0.99 / 1	how to weight matters
AJRY	$61.4 \rightarrow 45.3$	$0.81 \rightarrow 0.77$	J verdict flips, Δ falls

- J: how far are the moments from zero?
- Δ: how much of the variance do the moments explain?

Takeaways

1. Robust to misspecification: evaluate sensitivity and informativeness at the pseudo-true value.
2. Sensitivity rankings can reorder (BLP).
3. Weight choice in MD estimation is an efficiency–informativeness trade (BPP).
4. Reporting $\widehat{\Delta}_k$ is straightforward if you calculate misspecification-robust standard errors: it is free and answers what the J-test cannot (AJRY).

Appendix: Assumption 1 (regularity)

ASSUMPTION 1 *(regularity)*

i. $\{X_i\}_{i=1}^n$ i.i.d.
ii. $\Theta \subset \mathbb{R}^p$ compact; unique interior minimizer θ_0 of the population objective.
iii. $g(X, \cdot)$ thrice continuously differentiable; $\mathbb{E}\,\mathbf{sup}_\theta \,\|g\|^2$, $\mathbb{E}\,\mathbf{sup}_\theta \,\|G\|^2 < \infty$, and uniform L^2 (L^1) bounds on second (third) derivatives.
iv. curvature matrices A and $A + B$ nonsingular.
v. (iterated GMM) the population updating map is a contraction at θ_0 $(\rho(-A^{-1}B) < 1)$.
vi. weight conditions: (a) one-step, $\widehat{W} \xrightarrow{p} W \succ 0$, $\sqrt{n}\ \mathbf{vec}(\widehat{W} - W) = n^{-1/2} \sum_i \omega_i + o_p(1)$; (b) parameter-dependent, $\widehat{\phi} \xrightarrow{p} \phi_0$ asymptotically linear, $W(\phi)$ smooth with uniform convergence of $\widehat{W}(\phi)$ and its derivative.

Return to:

Prop 1 Prop 2 Prop 3 Prop 4

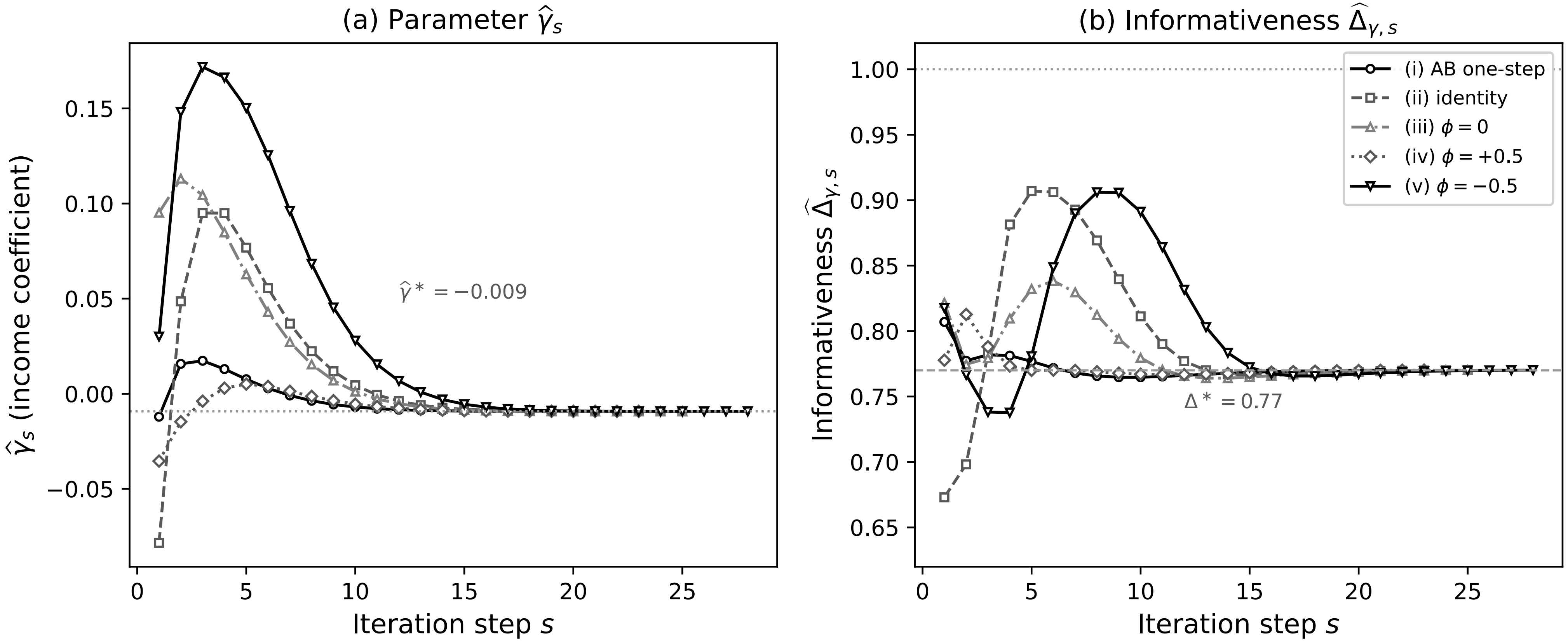
Appendix: Notation

- Primitives: $g = \mathbb{E}[g(X_i, \theta_0)]$, $\quad G = \mathbb{E}\left[\partial g(X_i, \theta_0)/\partial\theta'\right]$, $\quad W = \operatorname{plim} \widehat{W}$.
- Channels: $\nu_i = g(X_i, \theta_0) - g$, $\quad \gamma_i = \operatorname{vec}\left(G(X_i, \theta_0)' - G'\right)$, $\quad \omega_i =$ influence function of $\widehat{W}$, $\quad \psi^\phi =$ influence function of the first-step $\widehat{\phi}$.
- Curvature: $R = \mathbb{E}\left[\partial_{\theta'} \operatorname{vec}(G(X_i, \theta_0)')\right]$, $\quad H = (g'W \otimes I_p)R$, $\quad A = G'WG + H$.
- Iteration: $S = \partial_{\phi'} \operatorname{vec} W(\phi_0)$ (two-step) or $\partial_{\theta'} \operatorname{vec} W(\theta_0)$ (iterated), $\quad B = (g' \otimes G')S$.
- Coefficient maps: $M_\nu = -A^{-1}G'W$, $\quad M_\gamma = -A^{-1}(g'W \otimes I_p)$, $\quad M_\omega = -A^{-1}(g' \otimes G')$.

Return to:

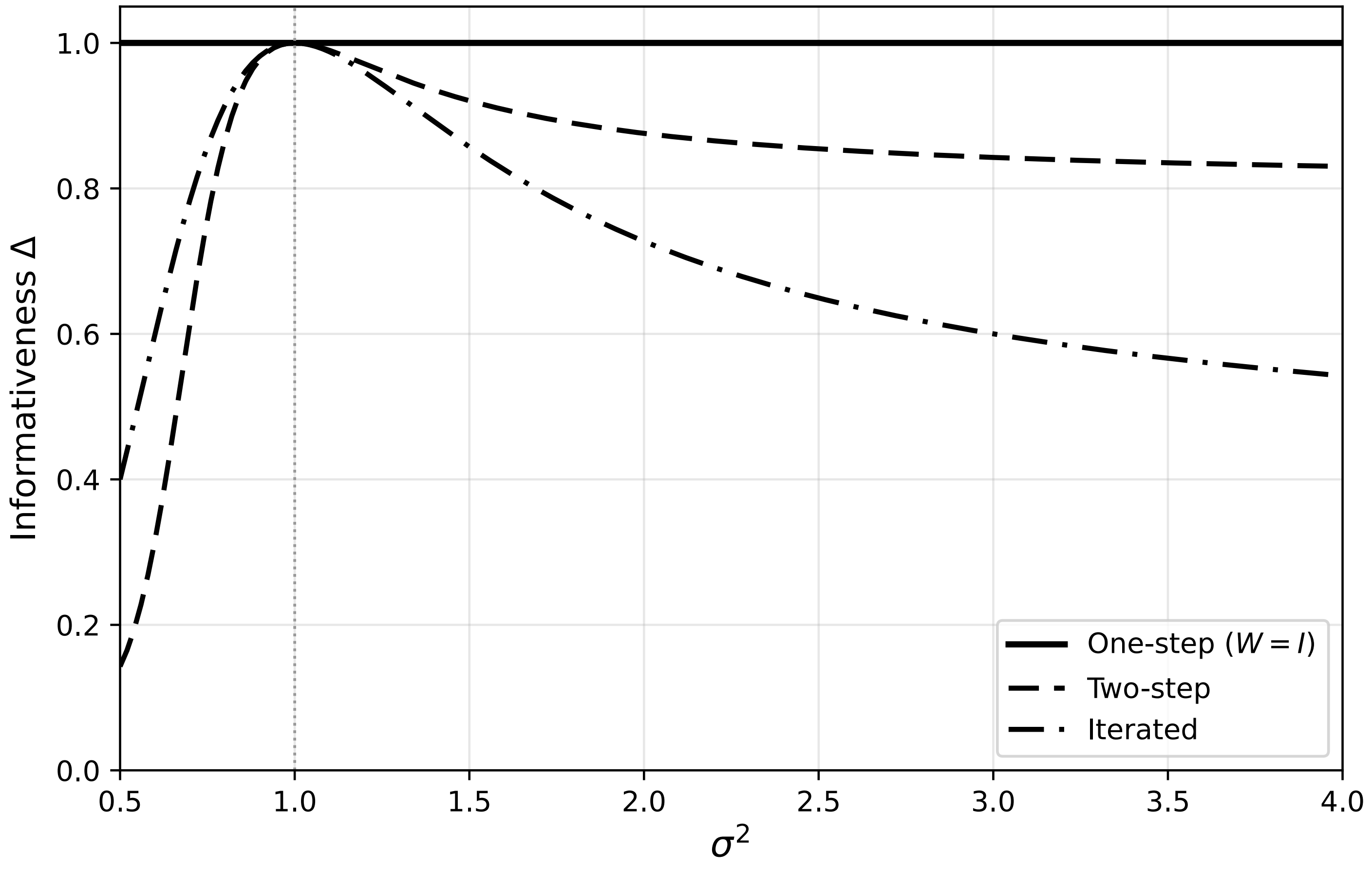
Prop 1 Prop 3

Appendix: AJRY multistart, common fixed point



$\widehat{\gamma}_s$ and $\widehat{\Delta}_{\gamma,s}$ reach the common fixed point from every start; the approach of Δ is non-monotone.

Appendix: Schennach closed form



Schennach (2007): normal mean with misspecified variance restriction; closed-form Δ for one-step, two-step, iterated, CUE.

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